Pes Planus Morphology and Foot Injury Incidence in Athletic and Non-Athletic Cohorts

[*1]Vidona W.B., [1]Eghaghe J., [2]Omoyibo C.A., [3]Dare E.I.

[1] Department of Anatomy, Ambrose Alli University Ekpoma, Edo State, Nigeria
[2] Department of Anatomy, University of Benin, Benin City, N*igeria.*
[3]Department of Anatomy, University of Lagos, Lagos State, Nigeria.

ABSTRACT

Pes planus has been implicated as a potential risk factor for lower limb injuries, although available evidence remains inconsistent. This cross-sectional study investigated the prevalence of pes planus and its association with foot injuries in professional athletes and recreational non-athletes. A sample of 169 participants (18 years and above) was stratified into two groups: professional athletes and recreational non-athletes. Stratified random sampling was employed to ensure that both groups are adequately represented. Data were collected using structured questionnaires, physical examinations to determine the prevalence, the occurrence and types of foot injuries, and socio-demographic characteristics. The data obtained was analyzed using SPSS version 24 statistical software. The findings revealed that 32% of the participants did not exhibit pes planus, 30.8% had no prior pes planus evaluation, and 36.7% were unaware of their pes planus status. A history of foot injury was reported by 52.7% of the participants, with sprains (24.3%) and fractures (20.1%) being the most frequently reported injury types. Chi-square analysis indicated no statistically significant association between the presence of pes planus and the occurrence of foot injuries ($\chi^2 = 0.331$, df = 2, p = 0.848). Independent samples t-tests on foot measurements (width, length, and arch height) showed no significant differences between participants with and without foot injuries. These results suggest that pes planus may not be a significant independent risk factor for foot injuries in this population. The study underscores the complexity of injury etiology and highlights the potential influence of other biomechanical and environmental factors.

Keywords: Pes planus, Foot injuries, Athletes, Non-athletes, Prevalence.

***Corresponding author.** Tel.: . **E-mail address:**vidonanem@gmail.com

1. INTRODUCTION

Pes planus, also known as flat foot, is a condition in which the arches of the feet collapse, causing the entire sole of the foot to come into complete or near-complete contact with the ground. This condition can present in varying degrees of severity especially the rigid pes planus characterized by the absence of an arch, even when the foot is not bearing weight, and is often associated with more severe structural deformities [1], often linked with more persistent discomfort and a higher likelihood of complications.

Foot health plays a critical role in both sports performance and daily life activities as the foot serves as the foundation for locomotion, functioning to distribute the forces generated. In the context of sports, where athletes are subjected to repetitive, high-impact movements, maintaining optimal foot structure and biomechanics is essential for preventing injuries and enhancing performance. A deviation from normal foot arch structure, as seen in pes planus, can lead to abnormal movement patterns, altered load distribution, and excessive stress on certain areas of the foot and lower extremities. Athletes with flat feet, particularly those with flexible pes planus are prone to overpronation, a condition in which the foot rolls excessively inward upon contact with the ground; which has been widely recognized as a major contributing factor to

various musculoskeletal injuries, including shin splints, plantar fasciitis, and knee pain [2]. The biomechanical effects of pes planus, especially in high-performance sports, highlight the need for tailored interventions, such as the use of orthotic insoles, to correct movement patterns and reduce injury risks.

Pes planus is not only a concern for professional athletes; it also affects non-professional individuals, particularly those who engage in recreational physical activities or have occupations that require prolonged standing or walking. In children, the condition is relatively common, but many outgrow it as their foot structure matures. However, for those who continue to exhibit pes planus into adulthood, the condition can lead to chronic discomfort and a heightened risk of injury. Among professional athletes, pes planus is less prevalent due to the rigorous physical demands of elite sports and the selective processes that often exclude individuals with significant structural foot deformities. However, when athletes with pes planus are involved in sports that place heavy demands on the lower extremities such as running, basketball, or soccer the risk of injury increases considerably [3]. Studies have shown that athletes with flat feet are more likely to suffer from overuse injuries, which can hinder their performance and lead to long-term health issues [4].

The prevalence of foot injuries, particularly in professional athletes, has become a growing concern in sports medicine. Common foot injuries among this population include plantar fasciitis, Achilles tendinitis, stress fractures, and ankle sprains which are often exacerbated in individuals with flat feet due to the abnormal distribution of forces across the foot during dynamic movements [5]. It is known that pes planus alters the alignment of the lower extremities, creating imbalances that can lead to compensatory mechanisms in other parts of the body, such as the knees, hips, and lower back. For non-professional individuals, pes planus can also lead to discomfort and a reduced quality of life, as activities like walking, standing for extended periods, and performing everyday tasks can become painful. This is particularly evident in individuals who are overweight or obese, as the added body weight places additional stress on the already compromised foot structure [6].

Given the widespread impact of pes planus on both athletic performance and daily functioning, addressing this condition through early intervention and proper management is crucial. The study of pes planus patterns and the prevalence of foot injuries among both professional athletes and non-professional individuals continues to be a significant area of research in sports medicine and orthopedics. Understanding the relationship between foot structure and injury risk is essential for developing effective preventive strategies and improving the overall health and performance of individuals affected by pes planus.

The aim of this study is to investigate the patterns of pes planus and the prevalence of foot injuries among professional athletes and non-professionals, providing comparative data on the two groups including its relationship on the severity and frequency of foot injuries between flat-footed individuals and those with normal arches in professional athletes and non-professionals. By advancing our understanding of the relationship between pes planus and foot injuries, the study will pave the way for future research and practical applications in sports medicine, podiatry, and public health.

2. MATERIALS AND METHODS

2.1. Research design: This study employs a cross-sectional research design to assess the prevalence of pes planus and associated foot injuries among athletes and non-athletes. The cross-sectional design is selected due to its efficiency in capturing data at a single point in time, allowing for the comparison of injury patterns and pes planus prevalence between the two groups as well as the frequency and types of foot injuries sustained by the participants.

2.2. Sample population: The target population for this study comprises two distinct groups: professional athletes and non-athletes. Athletes are defined as individuals who participate in professional or semi-professional sports, while non-athletes include individuals who engage in recreational sports or physical activities on an amateur basis. The sample was drawn from various sports clubs, and recreational sports leagues to ensure a representative mix of participants.

2.3. Ethical considerations: Ethical approval was obtained from the Health Research Ethics Committee of Ambrose Alli University, Ekpoma with an ethical approval number 018/25. Informed consent was sought from each participant before sample collections, upon educating them on the need for the research.

2.4. Sampling techniques and sample size: Stratified random sampling was employed to ensure that both groups are adequately represented. The populations were divided into two strata: athletes (professional sport stars) and non-athletes (non-professional). Within each stratum, participants were randomly selected to participate in the study giving a total of 169 participants. This technique ensures that the sample accurately reflects the diversity within each group and minimizes selection bias.

2.5.Study criteria: The inclusive criteria for this study are participants aged 18 years and above who are actively involved in sports activities, either as professional athletes or non-professional athletes, and are willing to provide informed consent. The exclusive criteria include individuals not actively participating in sports during the study period, those who refuse to participate, or those with incomplete data submissions.

2.6. Data collection methods: Data was collected using a combination of questionnaires and physical examinations.

2.7. Research instruments: The research instrument for this study was a structured questionnaire designed to collect self-reported data on demographics, sports participation, and foot injury history and perceived impact of pes planus on their performance. The questionnaire was pre-tested on a small sample to ensure clarity and reliability before administration. The questionnaire was divided into sections to address specific areas. The demographic information section gathers details on age, gender, educational level, and occupation. The sports participation section focuses on the type of sports participated in, the duration of sports involvement, and the frequency of sports activities. The foot injury history section includes questions on whether the participant has experienced foot injuries, the type of injuries such as sprains or fractures, the duration and frequency of injuries, and any previous treatments or medical interventions related to foot injuries.

The questionnaire also addresses foot structure and symptoms by determining the presence of pes planus, symptoms related to the foot arch such as pain or discomfort, and any corrective measures used, including orthotics or special footwear. This structured approach ensures that data collected is comprehensive and directly relevant to the study's objectives.

2.8. Physical examinations: Participants were examined for physical examinations conducted by qualified healthcare professionals to assess the presence of pes planus and other foot abnormalities. Standardized procedures for foot assessment, including visual inspection and the use of diagnostic tools such as foot arch indices was employed. Measurements was taken to evaluate the degree of pes planus and any associated patterns. All research instruments will undergo validation through a pilot study to ensure their reliability and accuracy in capturing relevant data.

2.9. Data analysis: Data analysis was done using statistical software to evaluate the prevalence of pes planus and the incidence of

foot injuries. Descriptive statistics, such as frequencies and percentages was used to summarize the data. Comparative analysis, including chi-square tests and t-tests, was employed to examine differences in injury patterns and pes planus prevalence betweenathletes and non-athletes.

3. RESULTS

The socio-demographic characteristics and associated variables of respondents participating in the study are shown in table 1. The total number of 169 respondents were anlysed. Regarding age distribution, 24 (14.2%) were between 18 and 20 years, 41 (24.3%) were between 21 and 23 years, 37 (21.9%) were between 24 and 26 years, 34 (20.1%) were between 27 and 29 years, and 32 (18.9%) were 30 years and above. Gender distribution revealed that 105 (62.1%) respondents were male, while 63 (37.3%) were female. Based on occupation, 61 (36.1%) were professional or semi-professional athletes, 86 (50.9%) were non-professional athletes, and 21 (12.4%) were classified as others.

In terms of sport type, 42 (24.9%) respondents participated in basketball, 39 (23.1%) in football, 43 (25.4%) in running/track and field, and 44 (26.0%) engaged in other unspecified sports. Sport frequency indicated that 40 (23.7%) respondents participated daily, 49 (29.0%) participated less than once a week, 37 (21.9%) participated two times a week, and 42 (24.9%) participated three times a week. pes planus diagnosis revealed that 54 (32.0%) respondents had no pes planus, 52 (30.8%) had never been evaluated, and 62 (36.7%) were unsure of their pes planus status.

For pain location, 37 (21.9%) respondents reported experiencing pain in the ankle, 35 (20.7%) in the arch, 28 (16.6%) in the heel, 27 (16.0%) in the toes, and 41 (24.3%) in other areas. Regarding foot injuries, 89 (52.7%) respondents reported having experienced a foot injury, while 79 (46.7%) reported no injuries. Injury frequency data showed that 43 (25.4%) respondents experienced injuries very often, 41 (24.3%) often, 47 (27.8%) occasionally, and 37 (21.9%) not at all. The type of injuries recorded included Achilles tendon injuries in 28 (16.6%) respondents, fractures in 34 (20.1%), plantar fasciitis in 26 (15.4%), sprains in 41 (24.3%), and other injury types in 39 (23.1%).

The descriptive statistics of foot measurements, including left foot length, right foot width, left foot width, right arch, and left arch, are presented in table 2. The total number of respondents for all parameters was 169. For left foot length, the minimum measurement recorded was 25.01 cm, the maximum was 27.60 cm, with a mean value of 26.84 cm and a standard deviation of 0.57. Right foot width ranged from 7.94 cm to 28.86 cm, with a mean of 10.16 cm and a standard deviation of 3.29. Similarly, left foot width had a minimum of 7.71 cm and a maximum of 29.35 cm, with a mean value of 10.45 cm and a standard deviation of 3.68. For the right arch, the minimum recorded measurement was 6.07 cm, while the maximum was 10.00 cm, with a mean value of 9.12 cm and a standard deviation of 0.67. The left arch had measurements ranging from 6.60 cm to 10.30 cm, with a mean of 9.37 cm and a standard deviation of 0.71.

The association between the presence of pes planus and the occurrence of foot injuries was analyzed using the Chi-square test, as shown in table 3a. The Chi-square test revealed no significant association between pes planus and foot injuries (Pearson Chi-square = 0.331, $p = 0.848$). The likelihood ratio test ($p = 0.848$) and linear-by-linear association test ($p = 0.818$) also showed no significant relationship. Based on 168 valid cases, with no cells having expected counts less than 5, the results suggest that pes planus does not significantly influence foot injury occurrence.

The strength of the association between the presence of pes planus and the occurrence of foot injuries was evaluated using symmetric measures, as presented in table 3b. The symmetric measures (Phi and Cramer's V =

0.044, p = 0.848) showed a very weak and insignificant association between pes planus and foot injuries, confirming the Chi-square analysis. Pearson's R (0.018, p = 0.819) and Spearman's correlation (0.017, p = 0.827) also revealed no significant linear or ordinal relationship. Based on 168 valid cases, the results indicate no meaningful association between pes planus and foot injuries.

Table 1. Descriptive statistics of demography, sport type, frequency, and pes planus diagnosis

Discription	Parameters	Frequency (N=169)	Percent (100)
Age	18 to 20	24	14.2
	21 to 23	41	24.3
	24 to 26	37	21.9
	27 to 29	34	20.1
	30 and above	32	18.9
Gender	Male	105	62.1
	Female	63	37.3
Occupation	Professional/Semi-Professional Athlete	61	36.1
	Non-Professional Athlete	86	50.9
	Others	21	12.4
Sport type	Basketball	42	24.9
	Football	39	23.1
	Running/Track and Field	43	25.4
	Others	44	26.0
Sport frequency	Daily	40	23.7
	Less than once a week	49	29.0
	2 times a week	37	21.9
	3 times a week	42	24.9
pes planus Diagnosis	No	54	32.0
	Never evaluated	52	30.8
	Not sure	62	36.7
Pain Location	Ankle	37	21.9
	Arch	35	20.7
	Heel	28	16.6
	Toes	27	16.0
	Others	41	24.3
Foot Injury	Yes	89	52.7
	No	79	46.7
Injury Frequency	Very often	43	25.4
	Often	41	24.3
	Occasionally	47	27.8
	Not at all	37	21.9
Injury Type	Achilles tendon injury	28	16.6
	Fracture	34	20.1
	Plantar fasciitis	26	15.4
	Sprain	41	24.3
	Others	39	23.1

Table 2: Descriptive statistics of left foot length, right foot width, left foot width, right arch and left arch

Parameters	N	Minimum	Maximum	Mean	Std. Deviation
Left Foot Length (cm)	169	25.01	27.60	26.84	0.57
Right Foot Width (cm)	169	7.94	28.86	10.16	3.29
Left Foot Width (cm)	169	7.71	29.35	10.45	3.68
Right Arch (cm)	169	6.07	10.00	9.12	0.67
Left Arch (cm)	169	6.60	10.30	9.37	0.71

Table 3a. Association between the presence of pes planus and the occurrence of foot injuries

Chi-Square Tests	Value	Df	Asymptotic Significance (2-sided)
Pearson Chi-Square	.331[a]	2	.848
Likelihood Ratio	.331	2	.848
Linear-by-Linear Association	.053	1	.818
N of Valid Cases	168		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 24.45.

Table 3b. Strength of the association between the presence of Pes planus and the occurrence of Foot Injuries

Symmetric Measures		Value	Asymptotic Standard Error[a]	Approximate T[b]	Approximate Significance
Nominal by Nominal	Phi	.044			.848
	Cramer's V	.044			.848
Interval by Interval	Pearson's R	.018	.077	.229	.819[c]
Ordinal by Ordinal	Spearman Correlation	.017	.077	.219	.827[c]
N of Valid Cases		168			

a. Not assuming the null hypothesis.
b. Using the asymptotic standard error assuming the null hypothesis.
c. Based on normal approximation.

The association between the presence of pes planus and the frequency of foot injuries among student athletes was examined using Chi-Square tests, as shown in table 4. The Chi-Square tests showed no significant association between pes planus and foot injury frequency. The Pearson Chi-Square value was 7.985 ($p = 0.239$), and the Likelihood Ratio was 8.129 ($p = 0.229$). The Linear-by-Linear Association test yielded a value of 1.080 ($p = 0.299$), indicating no meaningful trend. Based on 168 valid cases, all with expected counts above 5, these results suggest no significant correlation between pes planus and injury frequency among student athletes.

The variations in foot width, foot length, and foot arch by the occurrence of foot injury were analyzed using independent t-tests, as presented in table 5. he analysis of foot dimensions and injury occurrence using independent t-tests revealed no significant differences. Right foot length (RFL) was 26.71 ± 0.53 cm for injured and 26.59 ± 0.58 cm for non-injured individuals ($p = 0.310$). Left foot length (LFL) was 26.91 ± 0.53 cm for injured

and 26.78 ± 0.57 cm for non-injured individuals (p = 0.224). Right foot width (RFW) was 10.43 ± 3.75 cm for injured and 9.88 ± 2.70 cm for non-injured individuals (p = 0.089). Left foot width (LFW) was 10.78 ± 4.28 cm for injured and 10.12 ± 2.84 cm for non-injured individuals (p = 0.052). Right arch (RA) measured 9.14 ± 0.71 cm for injured and 9.12 ± 0.62 cm for non-injured individuals (p = 0.607), while left arch (LA) was 9.37 ± 0.78 cm for injured and 9.39 ± 0.59 cm for non-injured individuals (p = 0.165). Overall, no statistically significant differences were found between the groups

.

Table 4. Association/Correlation Between pes planus and the frequency of foot Injuries by Student Athletes

Chi-Square Tests	Value	Df	Asymptotic Significance (2-sided)
Pearson Chi-Square	7.985[a]	6	.239
Likelihood Ratio	8.129	6	.229
Linear-by-Linear Association	1.080	1	.299
N of Valid Cases	168		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 11.45.

Table 5. Variations in Foot Width, Foot Length, and Foot Arch by Foot Injury

Parameters	Yes	No	T-test	P-value	Inference
RFL	26.71±0.53	26.59±0.58	1.327	.310	NS
LFL	26.91±0.53	26.78±0.57	1.567	.224	NS
RFW	10.43±3.75	9.88±2.70	1.090	.089	NS
LFW	10.78±4.28	10.12±2.84	1.158	.052	NS
RA	9.14±0.71	9.12±0.62	0.203	.607	NS
LA	9.37±0.78	9.39±0.59	-0.151	.165	NS

RFL = Right Foot Length (cm), LFL = Left Foot Length (cm), RFW = Right Foot Width (cm), LFW = Left Foot Width (cm), RA = Right Arch (cm), LA = Left Arch (cm).

4. DISCUSSION

The findings of this study provide valuable insights into the prevalence of pes planus and its association with foot injuries among sport stars and non-professional athletes. The socio-demographic analysis in table 1 revealed a diverse participant pool with a balanced representation of professional and non-professional athletes, allowing for comprehensive comparisons. A significant proportion of participants were either unsure of their pes planus status (36.7%) or had never been evaluated (30.8%), highlighting a gap in awareness and diagnosis. This finding aligns with the report by Bourdet *et al*. [7], who noted that pes planus often goes undiagnosed, particularly in non-professional settings where routine biomechanical assessments are less common. Despite this lack of awareness in pes planus was not significantly associated with the occurrence or frequency of foot injuries, as indicated by the Chi-square analysis ($p > 0.05$) in table 3a. The symmetric measures in table 3b further confirmed the weak and insignificant relationship between pes planus and foot injuries ($p = 0.848$). These findings corroborate the study by Tong *et al*. [8], which concluded that pes planus alone may not be a sole predictor of injury but one of several contributing factors.

Foot injuries were prevalent among the participants, with 52.7% of respondents

reporting at least one injury, as shown in table 1. Sprains (24.3%), fractures (20.1%), and Achilles tendon injuries (16.6%) were the most common types, consistent with the observations of Fong *et al*. [9], who identified similar injury patterns among athletes. However, the Chi-square tests in table 4 showed no significant association between pes planus and injury frequency ($p = 0.239$), suggesting that pes planus might not independently influence the likelihood of recurrent injuries.

Descriptive statistics of foot measurements provided in table 2 revealed substantial variability in foot width, length, and arch dimensions among participants. However, the independent t-tests in table 5 indicated no significant differences in these measurements between participants with and without injuries. For instance, right foot length was 26.71 ± 0.53 cm for injured individuals and 26.59 ± 0.58 cm for non-injured individuals ($p = 0.310$), while left foot width showed a marginally non-significant difference ($p = 0.052$). These findings align with the conclusions of Cain *et al*. [10], who reported that anatomical variations like pes planus might not directly correlate with injury risk but interact with external factors such as surface type and training regimen.

The absence of statistically significant associations between pes planus and injury occurrence or frequency (Chi-square $p = 0.848$ in table 3a) reinforces the idea that pes planus alone is not a decisive factor in determining injury risk. Symmetric measures and correlation coefficients further supported these findings by indicating weak and insignificant relationships. These results corroborate the findings of Tarantino and Brancaccio [11], who emphasized that injury causation is multifactorial, with extrinsic factors such as sports participation intensity playing a more prominent role than intrinsic factors like pes planus.

5. CONCLUSION

This study found no significant association between pes planus and the occurrence or frequency of foot injuries among student athletes. Similarly, foot dimensions such as foot length, foot width, and arch height were not significantly related to injury risk. These findings suggest that pes planus alone may not serve as a definitive risk factor for foot injuries in athletic populations. However, the observed prevalence of non-specific lower limb discomfort in participants with pes planus highlights the need for further investigation into its long-term implications on musculoskeletal health. pes planus should be assessed in conjunction with other biomechanical and external factors, such as gait mechanics, footwear choices, and activity type, when developing injury prevention strategies.

Funding Statement: This study received no external funding. All expenses, including procurement of materials and laboratory analyses, were personally financed by the researchers.

Conflict of Interest: The authors declare that there are no conflicts of interest regarding the publication of this manuscript.

REFERENCES

1. Staheli L.T., Chew D.E., Corbett M. The longitudinal arch: A study of military recruits. *The Journal of Bone and Joint Surgery. American Volume*, 1987; 69(3), 426-8.
2. Wright WG., Ivanenko YP., Gurfinkel VS. Foot anatomy specialization for postural sensation and control. *Journal of Neurophysiology*, 2012; 107(5), 1513–1521.
3. Poppler LH., Moran SL. Acute Distal Radioulnar Joint Instability: Evaluation and Treatment. *Hand Clinics*, 2020; 36(4),429–441.
4. Mosca VS. Flexible Flatfoot in Children and Adolescents. *Journal of Children's Orthopaedics*, 2010; 4(2), 107–121.

5. Abegaz BA., Awoke DG. Factors affecting foot arch development in Northern Ethiopia. *Anatomy*, 2017; 11(1), 26–29.
6. Ibikunle PO., Ikekwem EC. Prevalence of pes planus. *Nigerian Journal of Medicine and Rehabilitation*, 2017; 19(1), 1–19.
7. Bourdet C., Seringe R., Adamsbaum C., Glorion C., Wicart P. Flatfoot in children and adolescents: Analysis of imaging findings and therapeutic implications. *Orthopaedics & Traumatology: Surgery & Research, 2013; 99*(1),80–87.
8. Tong JWK., Kong PW. 43Association between foot type and lower extremity injuries: Systematic literature review with meta-analysis. *Journal of Orthopaedic & Sports Physical Therapy,* 2013; (10), 700–714.
9. Fong DTP, Hong Y, Chan LK, Yung PSH, Chan KM. A systematic review on ankle injury and ankle sprain in sports. *Sports Medicine,* 2007; *37*(1), 73–94.
10. Cain LE., Nicholson LL., Adams RD., Burns J. Foot morphology and foot/ankle injury in indoor football. *Journal of Science and Medicine in Sport,* 2007; *10* (5), 311–319.
11. Tarantino D., Brancaccio P. Sports injury prevention: A concise review. *Pharmacologyonline, Special Issue 2019;* (1), 1–11.